

Decision Support

A Bayesian prioritization procedure for AHP-group decision making [☆]

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Abstract

This paper proposes a new prioritization procedure for Analytic Hierarchy Process Group Decision Making (AHP-GDM). Unlike the methods normally employed in AHP-GDM, this process does not require intermediate filters for the actors' initial judgements. The procedure is based on a Bayesian analysis of the problem and in general, it provides more efficient estimates than the techniques conventionally applied in the literature for AHP-GDM: aggregation of individual judgements (AIJ) and aggregation of individual priorities (AIP). The proposed procedure naturally extends to the analysis of incomplete and/or imprecise pairwise comparison matrices, enhancing realism, practicality and scope. The methodology has been illustrated by the analysis of a case study.

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1. Introduction

In view of the globalisation of the Knowledge-Based Society, on the one hand, and high levels of individual specialization, on the other, it is essential that the paradigms used in the scientific resolution of highly complex problems in the field of the social sciences should perform well in both individual and group decision making (Saaty, 1996).

One of the most widely used multi-criteria techniques in decision making processes involving multiple actors, scenarios and criteria is the Analytic Hierarchy Process (AHP), first proposed by Thomas L. Saaty in the 1970s (Saaty, 1980). It is also one of the methods that best captures changes in philosophy (from mechanistic reductionism to evolutionist holism), methodology (from the search for truth to the search for

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knowledge) and technology (communication networks) that took place in the latter years of the 20th century. This technique, like the majority of the multi-criteria approaches contained in the literature, has not escaped criticism (Belton and Gear, 1983, 1985; Barzilai and Golany, 1994; Barzilai, 1997). One way of mitigating the limitations of the method, at least in part, is to apply AHP through the prism of *cognitive constructivism* that characterises the *multi-criteria procedural rationality paradigm* (Moreno-Jiménez et al., 1999). This approach seeks to extract knowledge (behaviour patterns, trends, critical points, etc.) about the decision making process for use in the negotiation processes between the actors involved in the resolution of the problem.

In a local (unicriterion) context and assuming a scenario with multiple actors and a common hierarchy, the prioritization methods conventionally applied in AHP-GDM (Saaty, 1989; Ramanathan and Ganesh, 1994; Forman and Peniwati, 1998) use filters to “reduce” initial judgements, although this narrows their full range. The two most widely employed methods to obtain group priorities in AHP are: (i) the *aggregation of individual judgements* (AIJ) and (ii) the *aggregation of individual priorities* (AIP). In the first case (AIJ), a judgement matrix is constructed for the group. Each entry in this reciprocal matrix of pairwise comparisons is obtained as the weighted geometric mean of individual judgements and the priorities for the alternatives compared are calculated on this basis using any prioritization procedure. In the second case (AIP), the local priorities of each individual are first calculated using any prioritization procedure, and the group priorities are then obtained on the basis of these scores using a synthesis procedure, the most commonly used is the weighted geometric mean.

In both cases, it is assumed that the pairwise comparison matrices containing the judgements expressed by the decision makers are complete and accurate. However, when the analysis involves values for intangible attributes, it is not really appropriate to take such values as precise, as has been the case until now. At the same time, incomplete matrices including empty positions often have to be used in large problems. It may also sometimes be preferable, at least temporarily, to ignore those judgements that include opposing positions in order to increase consensus among decision makers, keeping only those pairwise comparison matrix entries that provide a degree of collective consensus. In view of this, the approach employed in the AHP-GDM process must perform appropriately both for precise and imprecise matrices (judgement intervals) as well as for incomplete matrices (empty entries).

This paper addresses these problems from a Bayesian standpoint. Taking the kind of multiplicative model with log-normal errors that is commonly used in this field, we propose a *Bayesian prioritization procedure* (BPP) that takes in the two conventional procedures (AIJ and AIP) as particular cases. These three procedures are then compared from a statistical point of view.

In line with the cognitive constructivism that characterises the multi-criteria procedural rationality paradigm (Moreno-Jiménez, 1997; Moreno-Jiménez et al., 1999, 2001), this new procedure allows us to treat the information provided by the actors on the judgements contained in the pairwise comparison matrices (which may be incomplete or imprecise) directly and without prior reduction, that is to say, without a previous synthesis of the individual elicited judgements. This provides greater realism and flexibility in the prioritization process.

The objective of the paper is not so much to find the “best” judgement aggregation procedure, which would be difficult in absolute terms, but to identify the knowledge provided by each prioritization process. This will improve our knowledge of the decisional process followed, in such a way that this knowledge can be exploited in subsequent negotiation processes. The paper is structured as follows. Section 2 briefly reviews the relevant theoretical background (AHP-group decision making and the Bayesian approach). Section 3 presents the Bayesian prioritization procedure put forward in the paper. Section 4 then goes on to apply the procedure to the treatment of incomplete and imprecise pairwise comparison matrices. Section 5 illustrates the proposed methodology using a practical case study. Finally, Section 6 concludes with a description of the main results obtained from the study.

2. Background

2.1. AHP group decision making (AHP-GDM)

The Analytic Hierarchical Process (AHP) is one of the methodological approaches that may be applied to resolve highly complex decision making problems involving multiple scenarios, criteria and actors. Proposed

in the 1970s by Thomas L. Saaty (Saaty, 1977, 1980), it constructs a ratio scale associated with the priorities for the various items compared. In his initial formulation, Conventional AHP, Saaty proposed a four-step methodology comprising modelling, valuation, prioritization and synthesis.

At the modelling stage, a hierarchy representing relevant aspects of the problem (criteria, sub-criteria, attributes and alternatives) is constructed. The goal or mission concerned in the problem is placed at the top of this hierarchy. Other relevant aspects (criteria, sub-criteria, attributes, etc.) are placed in the remaining levels. The preferences of the decision maker are included in the evaluation stage by means of reciprocal pairwise comparison matrices using judgements based on the fundamental scale proposed by Saaty. *Local priorities* for each element in the hierarchy are estimated in the prioritization stage. *Global priorities* for each element in the hierarchy, in particular the attributes included in the last but one level, are then estimated using the principle of hierarchical composition. Finally, the *total or final priorities* for the alternatives compared are obtained by synthesising the global priorities for the alternatives using one or other aggregation procedure (usually the additive procedure). Various authors have proposed alternatives to the scale utilised in Conventional AHP to evaluate judgements (Lootsma, 1993; Barzilai, 1997; Finan and Hurley, 1999), and for the prioritization and synthesis procedures applied (Saaty, 1980; Crawford and Williams, 1985; Bo, 1991; Aguarón et al., 1995; Aguarón and Moreno-Jiménez, 2000).

In contrast to other multi-criteria techniques, AHP permits an assessment of the degree of inconsistency present in the judgements expressed by the decision maker in the reciprocal positive pairwise comparison matrix. The two most widely used procedures in the literature are the *Consistency Ratio* (Saaty, 1980) and the *Geometric Consistency Index* (Crawford and Williams, 1985; Genest and Rivest, 1994; Aguarón and Moreno-Jiménez, 2003) used for the right eigenvector prioritization method (EGVM) and the row geometric mean method (RGMM), respectively.

AHP's flexibility and adaptability mean that the methodology can be used in decision making processes involving multiple actors. Moreno-Jiménez et al. (2002) distinguished between three multiple actor decision making situations, namely (i) group decision making, (ii) negotiated decision making and (iii) systematic decision making. In the first scenario, individuals work together in pursuit of a common goal. In the second, each individual resolves the problem separately and all of them then seek zones of agreement and disagreement between the actors. Finally, in the third case, each individual is allowed to act independently and the principle of tolerance is then invoked to establish the manner in which the different positions can be integrated.

The AHP literature describes two different ways of approaching group decision making with a view to obtaining group priorities (Saaty, 1989; Ramanathan and Ganesh, 1994; Forman and Peniwati, 1998). These are: (i) to work with *individual judgements* (AIJ) and (ii) to work with *individual priorities* (AIP). In the AIJ procedure, a new judgement matrix for the group as a whole is constructed on the basis of individual judgements, with the group's priorities being drawn from this new matrix. The three most commonly used methods to determine the entries in the group judgement matrix are *consensus*, *voting* and the *aggregation of judgements*. In AIP, the group's priorities are established on the basis of individual priorities using an aggregation method. With respect to both procedures, the most widely used aggregation technique is the weighted geometric mean.

2.2. The Bayesian approach in AHP

Despite the relative scarcity of references to Bayesian analysis in the AHP literature, the application of this approach has become more commonplace; particularly since the appearance of MCMC methods (see Robert and Casella, 1999 for a full discussion). These methods allow for the estimation and comparison of highly flexible and realistic hierarchical Bayesian models, as well as the treatment of incomplete and imprecise information using data augmentation techniques (Tanner and Wong, 1987).

Bayesian treatment provides a natural framework for the application of both inference processes regarding the parameters of a given model, as well as of model selection itself, incorporating the associated uncertainty (Bernardo and Smith, 1994) which adds to the rigor and realism of the analysis. Alho et al. (1996), were among the first to apply this approach to AHP, using a multiplicative model to break the error term down into three components that represent the variation between individuals, inconsistency in judgements and a residual error.

Alho and Kangas (1997) developed the previous results to include the case of a hierarchy and provided a Bayesian extension of the regression techniques employed in the earlier study. In particular, these authors used two techniques, the delta method and the simulation of the posterior distribution on the basis of non-informative prior distributions, to estimate the uncertainty associated with the estimated priorities.

The Bayesian approach also makes it possible to consider another kind of information issued by decision makers. For example, Basak (1998) utilises a model based on Dirichlet processes or mixtures of beta and normal distributions, and applies MCMC methods to calculate the posterior distributions of judgements. Using the geometric mean method, this author was also able to calculate the vector of priorities and the most likely rankings. Based on this information, non-parametric tests were then applied to search for significant differences between the alternatives. In this context, Basak (2001) assumes that experts classify judgements a_{ij} in k mutually exclusive intervals, assigning a probability that a given judgement will fall within each interval. On the basis of this information, he calculates the posterior distributions of judgements and estimates the group priority vectors utilising the minimum χ^2 method.

Against this background, the balance of this paper will be dedicated to presenting a new Bayesian procedure (BPP) for the aggregation of judgements and the estimation of priorities. This new method will then be compared with the two procedures conventionally used in AHP-GDM (AIJ and AIP).

3. A Bayesian prioritization procedure for AHP-GDM

Assuming a local (single criterion) context, let $\mathbf{G} = \{D_1, \dots, D_r\}$ $r \geq 2$ be a group of r decision makers expressing r reciprocal judgement matrices $\{\mathbf{A}^{(k)}, k = 1, \dots, r\}$ corresponding to pairwise comparisons with regard to the criterion considered for a set of n alternatives $\{A_1, \dots, A_n\}$, where $\mathbf{A}^{(k)} = (a_{ij}^{(k)})$ is a positive square matrix $(n \times n)$ which validates $a_{ii}^{(k)} = 1, a_{ji}^{(k)} = \frac{1}{a_{ij}^{(k)}} > 0$ for $i, j = 1, \dots, n$.

The judgements $a_{ij}^{(k)}$ represent the relative importance to decision maker D_k of alternative i compared to alternative j , according to the basic scale proposed by Saaty (1980). Let $\{v_1^G, \dots, v_n^G\}$ ($v_i^G \geq 0$) be the group's (unnormalised) priorities for the alternatives obtained by means of any of the usual prioritization methods, and let $w_1^G, \dots, w_n^G$ be their normalised distribution values: $w_i^G = v_i^G / \sum_{j=1}^n v_j^G$.

3.1. The model

In this paper we adopt a multiplicative model with log-normal errors, in line with those widely employed in the literature (Ramsay, 1977; De Jong, 1984; Crawford and Williams, 1985; Fichtner, 1986; Genest and Rivest, 1994; Alho and Kangas, 1997; Aguarón and Moreno-Jiménez, 2000; Laineinen and Hämäläinen, 2003). Assuming that the decision makers have expressed all possible judgements, this model is given by

$$a_{ij}^{(k)} = \frac{v_i^G}{v_j^G} e_{ij}^{(k)}, \quad i, j = 1, \dots, n, \quad k = 1, \dots, r,$$

with $e_{ij}^{(k)} \sim LN(0, \sigma^{(k)2})$ where $LN(\mu, \sigma^2)$ denotes the log-normal distribution of parameters μ and σ^2 , $e_{ji}^{(k)} = \frac{1}{e_{ij}^{(k)}}$ for $1 \leq i < j \leq n$, $e_{ii}^{(k)} = 1$ for $i = 1, \dots, n$, $k = 1, \dots, r$ and where $\mathbf{v}^G = (v_1^G, \dots, v_n^G)$ is the vector of the group's priorities.

Hence, it is assumed that consensus exists among the decision makers with regard to the priorities for each alternative. However, the degree of inconsistency for each decision maker ($\sigma^{(k)2}$), which is assumed to be known and below the threshold established in the problem, may differ.

By eliminating reciprocal judgements ($a_{ij}, i \geq j$) and degenerate situations, and taking logarithms, a regression model with normal errors is obtained

$$y_{ij}^{(k)} = \mu_i^G - \mu_j^G + \varepsilon_{ij}^{(k)}, \quad i = 1, \dots, n-1, \quad j = i+1, \dots, n, \quad k = 1, \dots, r, \quad (3.1)$$

where $y_{ij}^{(k)} = \log(a_{ij}^{(k)})$, $\mu_i^G = \log(v_i^G)$ and $\varepsilon_{ij}^{(k)} = \log(e_{ij}^{(k)}) \sim N(0, \sigma^{(k)2})$ are independent for $k = 1, \dots, r$. In order to avoid problems of identification, the alternative A_n is established as the benchmark alternative ($\mu_n = 0 \iff v_n = 1$).

In matrix notation, model (3.1) could be expressed as

$$\mathbf{y}^{(k)} = \mathbf{X}\boldsymbol{\mu}^G + \boldsymbol{\varepsilon}^{(k)} \quad \text{with } \boldsymbol{\varepsilon}^{(k)} \sim N_J(0, \sigma^{(k)2}\mathbf{I}_J), \quad (3.2)$$

where $J = n(n-1)/2$, $\mathbf{y}^{(k)} = (y_{12}^{(k)}, y_{13}^{(k)}, \dots, y_{n-1,n}^{(k)})'$ $k = 1, \dots, r$; $\boldsymbol{\mu}^G = (\mu_1^G, \dots, \mu_{n-1}^G)'$, $\boldsymbol{\varepsilon}^{(k)} = (\varepsilon_{12}^{(k)}, \varepsilon_{13}^{(k)}, \dots, \varepsilon_{n-1,n}^{(k)})'$ and $\mathbf{X}_{(J \times (n-1))} = (x_{pq})$ with $x_{pi} = 1$, $x_{pj} = -1$ and $x_{p\lambda} = 0$ if $\lambda \neq i, j, \lambda = 1, \dots, n-1$ and $p = \frac{(2n-i)}{2}(i-1) + (j-i)$ with $1 \leq i < j < n$, $x_{pi} = 1$ and $x_{p\lambda} = 0$ if $\lambda \neq i, \lambda = 1, \dots, n-1$ and $p = \frac{(2n-i)}{2}(i-1) + (n-i)$.

3.2. Bayesian prioritization in AHP-GDM

In the following discussion $[X]$ will denote the density of the random variable X and $[Y|X]$ the density of the conditional distribution $Y|X$. The prioritization procedure proposed for complete and precise matrices is based on the analysis of model (3.1) from a non-informative Bayesian standpoint.

Theorem 1. Taking a constant non-informative distribution as the prior distribution for the vector of log-priorities $\boldsymbol{\mu}^G$, the posterior distribution of $\boldsymbol{\mu}^G$ for complete and precise information is given by

$$\boldsymbol{\mu}^G | \mathbf{y} \sim N_{n-1}(\hat{\boldsymbol{\mu}}_B, \hat{\boldsymbol{\Sigma}}_B), \quad (3.3)$$

where

$$\hat{\boldsymbol{\mu}}_B = \frac{\sum_{k=1}^r \tau^{(k)} \hat{\boldsymbol{\mu}}^{(k)}}{\sum_{k=1}^r \tau^{(k)}}, \quad \hat{\boldsymbol{\Sigma}}_B = \left(\sum_{k=1}^r \tau^{(k)} \right)^{-1} (\mathbf{X}'\mathbf{X})^{-1} = \left(\sum_{k=1}^r \tau^{(k)} \right)^{-1} \begin{pmatrix} 2/n & 1/n & \dots & 1/n \\ 1/n & 2/n & \dots & 1/n \\ \dots & \dots & \dots & \dots \\ 1/n & 1/n & \dots & 2/n \end{pmatrix},$$

$\tau^{(k)} = 1/\sigma^{(k)2}$ and $\mathbf{y} = (\mathbf{y}^{(1)'}, \mathbf{y}^{(2)'}, \dots, \mathbf{y}^{(r)'})'$. In particular, the Bayes estimator of $\boldsymbol{\mu}^G$ under quadratic loss is given by $\hat{\boldsymbol{\mu}}_B$.

Proof. See Appendix A. $\square$

3.3. Aggregation of individual judgements (AIJ)

In the judgement aggregation process, a matrix of aggregated judgements $\mathbf{A}^G = (a_{ij}^G)$ is constructed using the geometric mean ($a_{ij}^G = \sqrt[r]{\prod_{k=1}^r a_{ij}^{(k)}}$). Under assumptions (3.1), we have

$$y_{ij}^G = \log(a_{ij}^G) = \frac{1}{r} \sum_{k=1}^r y_{ij}^{(k)} = \mu_i^G - \mu_j^G + \frac{1}{r} \sum_{k=1}^r \varepsilon_{ij}^{(k)} = \mu_i^G - \mu_j^G + \bar{\varepsilon}_{ij}^G \quad (3.4)$$

with $\bar{\varepsilon}_{ij}^G \sim N(0, \frac{1}{r} \sum_{k=1}^r \sigma^{(k)2}) = N(0, \bar{\sigma}^2)$, $i = 1, \dots, n-1$, $j = i+1, \dots, n$ and $k = 1, \dots, r$.

In this case, the maximum likelihood estimator of the vector of the group's log priorities, $\boldsymbol{\mu}^G$, will be obtained by the arithmetic mean estimator given by

$$\hat{\boldsymbol{\mu}}_{AIJ} = (\hat{\mu}_{AIJ,1}, \dots, \hat{\mu}_{AIJ,n-1})' \quad \text{with } \hat{\mu}_{AIJ,i} = \bar{y}_i^G - \bar{y}_n^G \text{ and } \frac{1}{n} \sum_{j=1}^G y_{ij}^G, \quad i = 1, \dots, n-1.$$

Let us note here that $\hat{\boldsymbol{\mu}}_{AIJ}$ is also the vector of the posterior mean distribution of this parameter when a constant non-informative prior distribution is adopted for $\boldsymbol{\mu}^G$ and the model considered is (3.4). Hence, it is a Bayesian estimator of $\boldsymbol{\mu}^G$ when a quadratic loss function is utilised. The following theorem compares the efficiency of estimators $\hat{\boldsymbol{\mu}}_B$ and $\hat{\boldsymbol{\mu}}_{AIJ}$.

Theorem 2. Under the assumptions of (3.1), the proposed Bayesian estimator for complete and precise matrices with known variance ($\hat{\boldsymbol{\mu}}_B$) is more efficient than the usual ($\hat{\boldsymbol{\mu}}_{AIJ}$) applied to aggregate judgements.

Proof. See Appendix A. $\square$

The relative efficiency of $\hat{\mu}_{AIJ}$ with respect to $\hat{\mu}_B$ is given – see expression (A.3) of Appendix A – by the ratio $\frac{\bar{\sigma}_A^2}{\bar{\sigma}_H^2} \geq 1$, where $\bar{\sigma}_A^2$ and $\bar{\sigma}_H^2$ represent the arithmetic and the harmonic mean, respectively, of the variances $\sigma^{(k)2}$, $k = 1, \dots, r$. This ratio reflects the loss of information in the group priorities estimation process caused by the aggregation of judgements. This loss will be minimal when $\sigma^{(k)2} = \sigma^2 \forall k = 1, \dots, r$, in which case, $\hat{\mu}_B = \hat{\mu}_{AIJ}$. In this sense, the more similar the inconsistency levels of the decision makers, the less will be the loss of information in the group priorities estimation process due to the judgement aggregation procedure.

However, if these levels of inconsistency are very different, then the loss of efficiency can be quite marked and the Bayesian estimator $\hat{\mu}_B$ could be much more efficient. This fact can be appreciated, for instance, if we consider the situation with $r = 2$. In this case,

$$\frac{\bar{\sigma}_A^2}{\bar{\sigma}_H^2} = \frac{1}{4} \left(2 + \frac{\sigma^{(1)2}}{\sigma^{(2)2}} + \frac{\sigma^{(2)2}}{\sigma^{(1)2}} \right) \rightarrow \infty \quad \text{when} \quad \max \left\{ \frac{\sigma^{(1)2}}{\sigma^{(2)2}}, \frac{\sigma^{(2)2}}{\sigma^{(1)2}} \right\} \rightarrow \infty.$$

3.4. Aggregation of individual priorities (AIP)

This process consists of aggregating the estimates of the priority vector for each decision maker D_k using a synthesis procedure. The most widely used procedure is the geometric mean, such that $\hat{\mu}_{AIP} = \frac{1}{r} \sum_{k=1}^r \hat{\mu}^{(k)}$, where $\hat{\mu}^{(k)} = (\hat{\mu}_1^{(k)}, \dots, \hat{\mu}_{n-1}^{(k)})'$ with $\hat{\mu}_i^{(k)} = \bar{y}_i^{(k)} - \bar{y}_n^{(k)}$ and where $\bar{y}_i^{(k)} = \frac{1}{n} \sum_{j=1}^n y_{ij}^{(k)}$ is the maximum likelihood and Bayesian estimation (when the loss function utilised is quadratic) of the log-priorities vector for decision maker D_k , $\mu^{(k)} = (\mu_1^{(k)}, \dots, \mu_{n-1}^{(k)})'$, obtained on the basis of the multiplicative model:

$$y_{ij}^{(k)} = \mu_i^{(k)} - \mu_j^{(k)} + v_{ij}^{(k)} \quad \text{with} \quad v_{ij}^{(k)} \sim N(0, \sigma^{(k)2}) \quad \text{for } 1 = i < j = n \text{ and } k = 1, \dots, r. \quad (3.5)$$

Under assumptions (3.5), the estimations obtained through the judgements and priorities aggregation procedures coincide and the results on efficiency described in the above section can be applied ($\hat{\mu}_B$ is more efficient than $\hat{\mu}_{AIP}$).

4. Extensions

In this section, the Bayesian treatment proposed in the preceding section is extended to the analysis of incomplete and/or imprecise pairwise comparison matrices. We also propose an algorithm, based on MCMC methods, which allows us to draw inferences about the group's priorities and the decision makers' levels of consistency.

4.1. Incomplete information

If the information provided by any of the decision makers is incomplete, meaning that not all of them express the $n(n-1)/2$ possible judgements in the reciprocal pairwise comparison matrix, the equations of model (3.2) will be given by

$$\mathbf{y}^{(k)} = \mathbf{X}^{(k)} \boldsymbol{\mu}^G + \boldsymbol{\varepsilon}^{(k)} \quad \text{with} \quad \boldsymbol{\varepsilon}^{(k)} \sim N_{J_k}(\mathbf{0}, \sigma^{(k)2} \mathbf{I}_{J_k}) \quad k = 1, \dots, r, \quad (4.1)$$

where J_k , $k = 1, \dots, r$, is the number of judgements issued by decision maker D_k and $\mathbf{X}^{(k)}$ is a matrix $J_k \times (n-1)$ similar to the matrix $\mathbf{X}$ defined earlier. In order to compare the different prioritization methods, we shall further assume that these matrices verify that $\det(\mathbf{X}^{(k)'} \mathbf{X}^{(k)}) > 0$, $k = 1, \dots, r$.

Model (4.1) may be expressed in matrix form as

$$\mathbf{y} = \mathbf{X}(\mathbf{1}_r \otimes \mathbf{I}_{n-1}) \boldsymbol{\mu}^G + \boldsymbol{\varepsilon} \quad \text{with} \quad \boldsymbol{\varepsilon} \sim N_j(\mathbf{0}, \mathbf{D}), \quad (4.2)$$

where $\mathbf{y} = (\mathbf{y}^{(1)'}, \mathbf{y}^{(2)'}, \dots, \mathbf{y}^{(r)'})' = (y_1, \dots, y_J)'$, $\mathbf{X} = \text{diag}(\mathbf{X}^{(1)}, \dots, \mathbf{X}^{(r)})$, $\boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}^{(1)'}, \boldsymbol{\varepsilon}^{(2)'}, \dots, \boldsymbol{\varepsilon}^{(r)'})'$, $\mathbf{1}_r$ is the $(r \times 1)$ vector $(1, \dots, 1)'$, $\mathbf{D} = \text{diag}(\sigma^{(1)2} \mathbf{I}_{J_1}, \dots, \sigma^{(r)2} \mathbf{I}_{J_r})$, $J = J_1 + \dots + J_r$ is the total number of judgements expressed by all decision makers and $\otimes$ denotes the Kronecker product.

Theorem 3. Taking a constant non-informative distribution as the prior distribution for the vector of log-priorities μ^G , the posterior distribution of μ^G for incomplete and precise information is given by

$$\mu|y \sim N_{n-1}(\hat{\mu}_B, \hat{\Sigma}_B), \quad (4.3)$$

where

$$\begin{aligned} \hat{\mu}_B &= \left(\sum_{k=1}^r \tau^{(k)} \mathbf{X}^{(k)'} \mathbf{X}^{(k)} \right)^{-1} \left(\sum_{k=1}^r \tau^{(k)} \mathbf{X}^{(k)'} \mathbf{y}^{(k)} \right) \\ &= \{ (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) (\mathbf{X}' \mathbf{D}^{-1} \mathbf{X}) (\mathbf{1}_r \otimes \mathbf{I}_{n-1}) \}^{-1} (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) (\mathbf{X}' \mathbf{D}^{-1} \mathbf{y}), \end{aligned} \quad (4.4)$$

$$\hat{\Sigma}_B = \left(\sum_{k=1}^r \tau^{(k)} \mathbf{X}^{(k)'} \mathbf{X}^{(k)} \right)^{-1}. \quad (4.5)$$

Proof. Similar to that for Theorem 1. $\square$

The estimator of μ^G obtained using the AIJ procedure will be constructed on the basis of the model

$$\mathbf{S}\mathbf{y} = \mathbf{S}\mathbf{X}(\mathbf{1}_r \otimes \mathbf{I}_{n-1})\mu^G + \mathbf{S}\boldsymbol{\varepsilon} \quad \text{with } \mathbf{S}\boldsymbol{\varepsilon} \sim N_j(\mathbf{0}, \mathbf{SDS}'), \quad (4.6)$$

where $\mathbf{S} = (s_{ij})$ is a $(J_{\text{dif}} \times J)$ matrix in which J_{dif} is the number of different judgements elicited by any of the r decision makers, $s_{ij} = \frac{1}{r_i}$ if $j \in \mathbf{M}_i = \{h \in \{1, \dots, J\} \text{ with } y_h \text{ being the } i\text{th of the different judgements}\}$, r_i is the cardinal of $\mathbf{M}_i$ reflecting the number of decision makers who have expressed the i th judgement and $s_{ij} = 0$ otherwise. The estimator μ^G will be given by

$$\hat{\mu}_{\text{AIJ}} = \{ (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) (\mathbf{X}' \mathbf{S}' \mathbf{S} \mathbf{X}) (\mathbf{1}_r \otimes \mathbf{I}_{n-1}) \}^{-1} (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) (\mathbf{X}' \mathbf{S}' \mathbf{S} \mathbf{y}). \quad (4.7)$$

The estimator of μ^G obtained by means of the AIP procedure will be given by

$$\hat{\mu}_{\text{AIP}} = \frac{1}{r} \sum_{k=1}^r \hat{\mu}^{(k)} = \frac{1}{r} \sum_{k=1}^r (\mathbf{X}^{(k)'} \mathbf{X}^{(k)})^{-1} (\mathbf{X}^{(k)'} \mathbf{y}^{(k)}) = \frac{1}{r} (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) (\mathbf{X}' \mathbf{X})^{-1} (\mathbf{X}' \mathbf{y}). \quad (4.8)$$

Remark 1

- (a) The three estimators of μ^G ($\hat{\mu}_B, \hat{\mu}_{\text{AIJ}}, \hat{\mu}_{\text{AIP}}$) are linear functions of the data ($\mathbf{y}$) given by $\mathbf{E}\mathbf{y}$, where matrix $\mathbf{E}$ verifies that $\mathbf{E}\mathbf{X}(\mathbf{1}_r \otimes \mathbf{I}_{n-1}) = \mathbf{I}_{n-1}$ and, therefore, the three estimators are unbiased.
- (b) Following elemental operations, it can be seen that if $\sigma^{(k)2} = \sigma^2 \forall k = 1, \dots, r$, then $\hat{\mu}_B = \hat{\mu}_{\text{AIP}}$. Moreover, if the $\mathbf{X}^{(k)}$ are the same for $k = 1, \dots, r$, then $\hat{\mu}_B = \hat{\mu}_{\text{AIJ}} = \hat{\mu}_{\text{AIP}}$. This remark extends the result of Section 3 for complete matrices.

Theorem 4. For incomplete and precise information, the Bayesian estimator ($\hat{\mu}_B$) obtained from (4.4) is more efficient than those ($\hat{\mu}_{\text{AIJ}}, \hat{\mu}_{\text{AIP}}$) provided by expressions (4.7) and (4.8) for the AIJ or AIP, respectively.

Proof. See Appendix A. $\square$

4.2. Imprecise information

Occasionally, the information provided by the decision maker with regard to some of the judgements $y_{ij}^{(k)}$ is imprecise and is given in the form of an interval judgement of the type $\left[\underline{y}_{ij}^{(k)}, \bar{y}_{ij}^{(k)} \right] (i, j, k) \in M$ where $M \subseteq \{1, \dots, n\} \times \{1, \dots, n\} \times \{1, \dots, r\}$ with $\underline{y}_{ij}^{(k)} = \log(\underline{a}_{ij}^{(k)})$, $\bar{y}_{ij}^{(k)} = \log(\bar{a}_{ij}^{(k)})$ and where $\left[\underline{a}_{ij}^{(k)}, \bar{a}_{ij}^{(k)} \right]$ is an interval within which the value of the judgement $a_{ij}^{(k)}$ would fall. In this situation it is not clear how to carry out the judgement and/or the priority aggregation processes described above. To the best of our knowledge, there is no universally accepted procedure in the literature to perform this aggregation in the absence of a precise value for $y_{ij}^{(k)}$.

From a Bayesian standpoint, however, this does not present a problem. It is sufficient to treat the values of the imprecise judgements $y_{ij}^{(k)}$ as additional parameters of the model, the values of which are restricted by the limits of the intervals.

In the majority of cases a conjugated analysis will not be possible, and it will therefore be necessary to employ approximate inferential methods such as the “data augmentation technique” originally proposed by Tanner and Wong (1987), which is the most widely used in the Bayesian literature (Robert and Casella, 1999; Van dyk and Meng, 2001). This process consists of obtaining the posterior distribution $[\mu^G, \mathbf{y}_{\text{imp}} | \mathbf{y}_{\text{pre}}, \mathbf{y}_{\text{imp}} \in C]$, where $\mathbf{y}_{\text{imp}}$ and $\mathbf{y}_{\text{pre}}$ denote the vector of imprecise and precise judgements, respectively, and C is the Cartesian product $\prod_{(i,j,k) \in M} [y_{ij}^{(k)}, \bar{y}_{ij}^{(k)}]$. The calculation of this distribution can be made using Gibbs sampling, employing, to that end, the full conditional distributions $[\mu^G | \mathbf{y}_{\text{pre}}, \mathbf{y}_{\text{imp}}]$ and $[\mathbf{y}_{\text{imp}} | \mu^G, \mathbf{y}_{\text{pre}}]$, where the latter is truncated in C . Both distributions have “closed forms” and, therefore, it is easier to sample them. Once having calculated the distributions, the posterior distribution of the vector of priorities $\mu^G | \mathbf{y}_{\text{pre}}, \mathbf{y}_{\text{imp}} \in C$ will be calculated by marginalisation. For example, if the model is given by the expression (4.2), the following algorithm will provide a sample of the posterior distribution of μ^G by means of the Gibbs sampling, which can be used to make inferences about the group’s priorities.

Algorithm. Let $\{\mathbf{y}_{\text{pre}}^{(k)}\}$ and $\{\mathbf{y}_{\text{imp}}^{(k)}\}$ be the vectors of the precise and imprecise judgements, respectively, expressed by each of the decision makers involved in the problem ($k = 1, \dots, r$).

Step 0 (Initialisation):

Assign initial values $\mathbf{y}_{\text{imp}}^{(k,0)}$ to the imprecise judgements ($\mathbf{y}_{\text{imp}}^{(k)}$) by applying simulation techniques to the interval judgements. Set $it = 1$ and repeat Steps 1 and 2 below n_{iter} times.

Step 1 (Sampling of $\mu^G | \mathbf{y}_{\text{pre}}, \mathbf{y}_{\text{imp}}$):

Draw $\mu^{G(it)}$ from $N_{n-1} \left(\hat{\mu}_B^{(it-1)}, \left(\sum_{k=1}^r \tau^{(k)} \mathbf{X}^{(k)'} \mathbf{X}^{(k)} \right)^{-1} \right)$ where

$$\hat{\mu}_B^{(it-1)} = \left(\sum_{k=1}^r \tau^{(k)} \mathbf{X}^{(k)'} \mathbf{X}^{(k)} \right)^{-1} \left(\sum_{k=1}^r \tau^{(k)} \mathbf{X}^{(k)'} \mathbf{y}^{(k,it-1)} \right)$$

and $\{\mathbf{y}^{(k,it-1)} = (\mathbf{y}_{\text{pre}}^{(k)}, \mathbf{y}_{\text{imp}}^{(k,it-1)})', k = 1, \dots, r\}$ are the “complete” judgements expressed by each decision maker in which the values assigned in the preceding iteration are included.

Step 2 (Sampling of $\mathbf{y}_{\text{imp}} | \mu^G, \mathbf{y}_{\text{pre}}$ truncated in C):

For each imprecise judgement $y_{ij}^{(k)}$, draw $y_{ij}^{(k,it)}$ from the normal distribution $N(\mu_i^{G(k,it)} - \mu_j^{G(k,it)}, \sigma^{(k)2})$ truncated at the interval $[y_{ij}^{(k)}, \bar{y}_{ij}^{(k)}]$. On the basis of these random imputations, construct the judgement vectors $\{\mathbf{y}_{\text{imp}}^{(k,it)}; k = 1, \dots, r\}$.

The convergence of the algorithm to the posterior distribution of μ^G follows from Lemma 7.2.1 of Robert and Casella, 1999). As a result of this algorithm, a sample $\{\mu^{G(it)}; it = 1, \dots, n_{\text{iter}}\}$ of this distribution is obtained. This sample can be used to make inferences about the log-priorities of the group (μ^G). For example, Bayesian credibility intervals of $100(1 - \alpha)\%$ can be obtained for each of the priorities $\{w_i^j; i = 1, \dots, n\}$ simply by calculating the $\alpha/2$ and $1 - \alpha/2$ quantiles of the sample $\{w_i^{G(it)}; it = n_{\text{iter}0} + 1, \dots, n_{\text{iter}}\}$ with $w_i^{(it)} = \exp[\mu_i^{G(it)}] / \sum_{j=1}^n \exp[\mu_j^{G(it)}]$, taking $\mu_n^{(it)} = 0 \forall it$, and where $n_{\text{iter}0}$ is the number of iterations necessary to achieve convergence.

4.3. Unknown variance

Up to this point, we have assumed that the variances of the error terms $\{\sigma^{(k)2}, k = 1, \dots, r\}$ were known. These variances represent the degree of inconsistency of each decision maker in expressing their judgements, and they are usually unknown. A possible solution in this case is to include variances as additional parameters in (3.1) and adopt a non-informative prior distribution for each one given by $[\tau^{(k)}] = 1/\tau^{(k)}, k = 1, \dots, r$.

The resulting posterior distribution is given by

$$\begin{aligned} [\boldsymbol{\mu}^G, \tau^{(1)}, \dots, \tau^{(r)} | \mathbf{y}] &\propto [\mathbf{y} | \boldsymbol{\mu}^G, \tau^{(1)}, \dots, \tau^{(r)}] [\boldsymbol{\mu}^G] \prod_{k=1}^r [\tau^{(k)}] \propto \prod_{k=1}^r [\mathbf{y}^{(k)} | \boldsymbol{\mu}^G, \tau^{(k)2}] [\boldsymbol{\mu}^G] \prod_{k=1}^r [\tau^{(k)}] \\ &\propto \prod_{k=1}^r (\tau^{(k)})^{\frac{J_k}{2}-1} \exp \left[-\frac{\tau^{(k)}}{2} (\mathbf{y}^{(k)} - \mathbf{X}^{(k)} \boldsymbol{\mu}^G)' (\mathbf{y}^{(k)} - \mathbf{X}^{(k)} \boldsymbol{\mu}^G) \right]. \end{aligned} \quad (4.9)$$

Since this is not a conjugated analysis, it is necessary to apply approximation techniques to calculate the posterior distribution. MCMC methods, specifically Gibbs sampling (Robert and Casella, 1999), may be used for this purpose to obtain a sample of the distribution (4.9), which is then used to make inferences about $\boldsymbol{\mu}^G$ and the parameters $\sigma^{(1)}, \dots, \sigma^{(r)}$ (see the algorithm in Appendix B).

Alternatively, and basically with the same conclusions, informative prior distributions for the parameters $\boldsymbol{\mu}^G$ and $\tau^{(1)}, \dots, \tau^{(r)}$ can be supposed (Altuzarra et al., 2006). However, this extension exceeds the objective of this paper in which we have considered a non-informative prior distribution, widely used in Bayesian analysis, and where our main aim is to carry out a “closed” comparison of the three estimators being analysed.

5. Case study

Let us consider a group decision analysis situation taken from Wang and Xu (1990). Further, let us suppose that six decision makers give the following comparison matrices with regard to the same decision problem, respectively:

$$\begin{aligned} A_1 &= \begin{bmatrix} 1 & 3 & 5 & 4 & 7 \\ 1/3 & 1 & 3 & 2 & 5 \\ 1/5 & 1/3 & 1 & 1/2 & 3 \\ 1/4 & 1/2 & 2 & 1 & 3 \\ 1/7 & 1/5 & 1/3 & 1/3 & 1 \end{bmatrix}, & A_2 &= \begin{bmatrix} 1 & 4 & 3 & 5 & 8 \\ 1/4 & 1 & 4 & 3 & 6 \\ 1/3 & 1/4 & 1 & 1 & 5 \\ 1/5 & 1/3 & 1 & 1 & 7 \\ 1/8 & 1/6 & 1/5 & 1/7 & 1 \end{bmatrix}, \\ A_3 &= \begin{bmatrix} 1 & 1/2 & 3 & 2 & 5 \\ 2 & 1 & 5 & 1 & 2 \\ 1/3 & 1/5 & 1 & 2 & 1/2 \\ 1/2 & 1 & 1/2 & 1 & 5 \\ 1/5 & 1/2 & 2 & 1/5 & 1 \end{bmatrix}, & A_4 &= \begin{bmatrix} 1 & 3 & 5 & 2 & 6 \\ 1/3 & 1 & 1 & 3 & 2 \\ 1/5 & 1 & 1 & 4 & 5 \\ 1/2 & 1/3 & 1/4 & 1 & 1/2 \\ 1/6 & 1/2 & 1/5 & 2 & 1 \end{bmatrix}, \\ A_5 &= \begin{bmatrix} 1 & 2 & 6 & 3 & 3 \\ 1/2 & 1 & 2 & 5 & 4 \\ 1/6 & 1/2 & 1 & 1/2 & 1 \\ 1/3 & 1/5 & 2 & 1 & 5 \\ 1/3 & 1/4 & 1 & 1/5 & 1 \end{bmatrix}, & A_6 &= \begin{bmatrix} 1 & 2 & 5 & 4 & 9 \\ 1/2 & 1 & 3 & 2 & 6 \\ 1/5 & 1/3 & 1 & 1 & 2 \\ 1/4 & 1/2 & 1 & 1 & 3 \\ 1/9 & 1/6 & 1/2 & 1/3 & 1 \end{bmatrix}. \end{aligned}$$

5.1. Bayesian analysis of individual pairwise comparison matrices

Tables 5.1 and 5.3 show the results obtained from a Bayesian analysis of the information provided by the above pairwise comparison matrices using model (3.5) for each decision maker separately. In each case, we have taken the prior distribution of the vector of parameters $(\boldsymbol{\mu}^{(k)}, \tau^{(k)})$, the non-informative distribution $[\boldsymbol{\mu}^{(k)}, \tau^{(k)}] \propto \frac{1}{\tau^{(k)}}$, $k = 1, \dots, 6$, with the result that the posterior distribution $(\boldsymbol{\mu}^{(k)}, \tau^{(k)}) | \mathbf{y}^{(k)}$ will be a Normal Gamma, such that:

$$\boldsymbol{\mu}^{(k)} | \tau^{(k)}, \mathbf{y}^{(k)} \sim N_{n-1}(\hat{\boldsymbol{\mu}}^{(k)}, (\tau^{(k)} \mathbf{X}' \mathbf{X})^{-1}) \quad (5.1)$$

$$\tau^{(k)} | \mathbf{y}^{(k)} \sim \text{Gamma} \left(\frac{10}{2}, \frac{(\mathbf{y}^{(k)} - \mathbf{X}' \hat{\boldsymbol{\mu}}^{(k)})' (\mathbf{y}^{(k)} - \mathbf{X}' \hat{\boldsymbol{\mu}}^{(k)})}{2} \right), \quad (5.2)$$

where $\text{Gamma}(\alpha, \beta)$ denotes the gamma distribution with shape parameter $\alpha > 0$ and inverse scale $\beta > 0$.

Table 5.1

Priorities and individual inconsistency levels (Bayesian credibility interval of 95% in parentheses)

	$w_1^{(k)}$	$w_2^{(k)}$	$w_3^{(k)}$	$w_4^{(k)}$	$w_5^{(k)}$	$\sigma^{(k)}$
D_1	0.491 (0.461,0.520)	0.232 (0.211,0.254)	0.092 (0.083,0.103)	0.138 (0.125,0.153)	0.046 (0.041,0.052)	0.259 (0.175,0.439)
D_2	0.480 (0.426,0.535)	0.249 (0.208,0.295)	0.117 (0.096,0.142)	0.120 (0.026,0.032)	0.032 (0.026,0.039)	0.481 (0.325,0.817)
D_3	0.301 (0.237,0.371)	0.318 (0.251,0.391)	0.102 (0.077,0.134)	0.183 (0.140,0.235)	0.092 (0.069,0.121)	0.699 (0.472,1.186)
D_3^a	0.365 (0.297,0.437)	0.253 (0.199,0.315)	0.130 (0.101,0.165)	0.180 (0.145,0.218)	0.069 (0.053,0.090)	0.552 (0.372,0.936)
D_4	0.451 (0.388,0.511)	0.183 (0.147,0.225)	0.210 (0.170,0.257)	0.073 (0.058,0.092)	0.081 (0.064,0.101)	0.565 (0.382,0.959)
D_4^a	0.456 (0.408,0.504)	0.185 (0.156,0.218)	0.174 (0.143,0.211)	0.119 (0.092,0.151)	0.063 (0.051,0.078)	0.445 (0.301,0.755)
D_5	0.407 (0.349,0.463)	0.290 (0.241,0.341)	0.084 (0.068,0.105)	0.147 (0.119,0.180)	0.071 (0.056,0.087)	0.526 (0.355,0.892)
D_6	0.475 (0.462,0.488)	0.261 (0.251,0.271)	0.098 (0.093,0.102)	0.120 (0.115,0.126)	0.046 (0.044,0.048)	0.112 (0.076,0.190)

^a Priorities and inconsistency levels for decision makers D_3 and D_4 omitting judgements $a_{12}^{(3)}$, $a_{35}^{(3)}$, $a_{34}^{(4)}$ and $a_{45}^{(4)}$.

Table 5.2

Matrices of domination between alternatives for each decision maker

D_1	A_1	A_2	A_3	A_4	A_5	D_2	A_1	A_2	A_3	A_4	A_5	D_3	A_1	A_2	A_3	A_4	A_5
A_1	0.00	1.00	1.00	1.00	1.00	A_1	0.00	1.00	1.00	1.00	1.00	A_1	0.00	0.38	1.00	0.99	1.00
A_2	0.00	0.00	1.00	1.00	1.00	A_2	0.00	0.00	1.00	1.00	1.00	A_2	0.62	0.00	1.00	0.99	1.00
A_3	0.00	0.00	0.00	0.00	1.00	A_3	0.00	0.00	0.00	0.43	1.00	A_3	0.00	0.00	0.00	0.00	0.72
A_4	0.00	0.00	1.00	0.00	1.00	A_4	0.00	0.00	0.57	0.00	1.00	A_4	0.01	0.01	1.00	0.00	1.00
A_5	0.00	0.00	0.00	0.00	0.00	A_5	0.00	0.00	0.00	0.00	0.00	A_5	0.00	0.00	0.28	0.00	0.00
D_4	A_1	A_2	A_3	A_4	A_5	D_5	A_1	A_2	A_3	A_4	A_5	D_6	A_1	A_2	A_3	A_4	A_5
A_1	0.00	1.00	1.00	1.00	1.00	A_1	0.00	0.99	1.00	1.00	1.00	A_1	0.00	1.00	1.00	1.00	1.00
A_2	0.00	0.00	0.17	1.00	1.00	A_2	0.01	0.00	1.00	1.00	1.00	A_2	0.00	0.00	1.00	1.00	1.00
A_3	0.00	0.83	0.00	1.00	1.00	A_3	0.00	0.00	0.00	0.00	0.90	A_3	0.00	0.00	0.00	0.00	1.00
A_4	0.00	0.00	0.00	0.00	0.26	A_4	0.00	0.00	1.00	0.00	1.00	A_4	0.00	0.00	1.00	0.00	1.00
A_5	0.00	0.00	0.00	0.74	0.00	A_5	0.00	0.00	0.10	0.00	0.00	A_5	0.00	0.00	0.00	0.00	0.00

Table 5.3

Most likely preference structures for each decision maker

	12345 ^a	12435	12354	12453	21435	21453	13245	13254
D_1	0.01	99.99	0.00	0.00	0.00	0.00	0.00	0.00
D_2	43.00	57.00	0.00	0.00	0.00	0.00	0.00	0.00
D_3	0.11	27.34	0.03	10.08	44.25	18.18	0.00	0.00
D_3^b	2.67	97.02	0.00	0.27	0.00	0.00	0.04	0.00
D_4	4.57	0.01	12.57	0.00	0.00	0.00	21.63	61.21
D_4^b	66.87	1.80	0.00	0.00	0.00	0.00	31.33	0.00
D_5	0.07	89.27	0.00	9.34	1.09	0.23	0.00	0.00
D_6	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00

^a Preference structure $A_1 > A_2 > A_3 > A_4 > A_5$ where $>$ means “preferred over”.^b Distribution of preference structures for decision makers D_3 and D_4 omitting judgements $a_{12}^{(3)}$, $a_{35}^{(3)}$, $a_{34}^{(4)}$ and $a_{45}^{(4)}$.

Table 5.1 reflects the posterior medians for the priorities of each decision maker $\hat{w}_i^{(k)} = \exp(\hat{\mu}_i^{(k)}) / \sum_{j=1}^5 \exp(\hat{\mu}_j^{(k)})$, $i = 1, \dots, 5$, $k = 1, \dots, 6$ and the inconsistency levels $\{\sigma^{(k)}; k = 1, \dots, 6\}$, as well as the 95% Bayesian credibility intervals constructed on the basis of the quantiles 2.5 and 97.5 of the posterior distributions. For its part, Table 5.2 shows the posterior probabilities of rank reversal $\{P[w_i^{(k)} > w_j^{(k)} | \mathbf{y}^{(k)}], 1 \leq i, j \leq 5, k = 1, \dots, 6\}$ for each decision maker and, finally, Table 5.3 shows the posterior probabilities of the most likely preference structures. These probabilities were obtained using the Monte Carlo method on the basis of the posterior distributions (5.1) and (5.2) with 10,000 simulations.

The highest priority alternative for decision maker D_3 was A_2 estimated at 0.3179, followed by A_1 (0.3005), A_4 (0.1828), A_3 (0.1018) and A_5 (0.0918) with a level of inconsistency equal to 0.6994 (see Table 5.1). The most likely preference structure is [21435] or $A_2 > A_1 > A_4 > A_3 > A_5$ (“ $>$ ” means “preferred over”) with a probability of 44.25% (see Table 5.3). In view of the high level of uncertainty accompanying the estimation of the

decision maker's priorities, mainly due to the high level of inconsistency, other structures also exhibit significant probabilities, such as [12435] (27.34%), [21453] (18.18%) and [12453] (10.08%). This is because the probabilities of rank reversal between alternatives A_1 and A_2 and A_3 and A_5 are fairly high, as it is estimated that $P(A_1 > A_2 | y^{(3)}) = 0.38$ and $P(A_5 > A_3 | y^{(3)}) = 0.28$ (see Table 5.2).

On the basis of the above three tables, we may deduce that decision makers D_1 and D_6 are the most consistent of the group, and their preference structure is given by [12435]. This structure is also supported by decision maker D_5 , but less strongly, since the probability of rank reversal between the alternatives A_3 and A_5 is not negligible (see Table 5.2). It is also supported, although to a lesser extent, by decision maker D_2 , who appears to be unsure of the order between alternatives A_3 and A_4 , with a probability of rank reversal between them at the 43% level. For these reasons, the inconsistency levels of these two decision makers are higher than for D_1 and D_6 .

The decision makers who are most clearly in disagreement with the aforementioned preference structure are D_3 and D_4 , who are also the most inconsistent of the six (see Table 5.1). As we have already remarked, decision maker D_3 opts for the ranking [21435], but not very clearly in view of the high probability of rank reversal between the alternatives A_1 and A_2 and A_3 and A_5 . Finally, decision maker D_4 is the most strongly in disagreement with the structure [12435], as shown by the low posterior probability assigned (0.01%), and opts for the structure [13254] (see Table 5.3). There is, however, some likelihood of rank reversal between alternatives A_2 and A_3 and A_5 and A_4 , as reflected in the not insignificant probabilities of the preference structures [13245] and [12354] (see Table 5.3).

5.2. Estimation of group priorities and preferences

Tables 5.4–5.7 show the results obtained using each of the prioritization methods described in this paper for the group priorities $\{w_i^G; i = 1, \dots, 5\}$ (Table 5.4), individual inconsistency levels (Table 5.5), the probability of rank reversal between the alternatives (Table 5.6) and the most likely preference structures (Table 5.7). In these tables, the Bayesian method employs the posterior distribution (4.9) to make the above inferences, applying the algorithm described in the Appendix over 10,000 iterations, the first 5000 being discarded. The inferences made by the AIJ and AIP methods are based on models (3.4) and (3.5), respectively.

Table 5.4
Group priorities estimated by each method (Bayesian credibility interval of 95% in parentheses)

	w_1	w_2	w_3	w_4	w_5
AIJ	0.439 (0.418,0.461)	0.258 (0.241,0.276)	0.114 (0.105,0.124)	0.129 (0.119,0.139)	0.059 (0.054,0.063)
AIP	0.439 (0.368,0.509)	0.257 (0.205,0.319)	0.114 (0.087,0.146)	0.129 (0.099,0.165)	0.058 (0.044,0.076)
Bayesian	0.479 (0.451,0.510)	0.259 (0.235,0.283)	0.097 (0.086,0.107)	0.119 (0.107,0.132)	0.045 (0.040,0.051)
AIJ ^a	0.455 (0.423,0.487)	0.239 (0.216,0.264)	0.112 (0.100,0.126)	0.143 (0.127,0.160)	0.050 (0.044,0.057)
AIP ^a	0.448 (0.383,0.512)	0.245 (0.198,0.300)	0.113 (0.090,0.143)	0.137 (0.108,0.173)	0.053 (0.041,0.068)
Bayesian ^a	0.479 (0.451,0.511)	0.259 (0.235,0.282)	0.097 (0.086,0.107)	0.119 (0.106,0.132)	0.045 (0.040,0.051)
Bayesian ^b	0.479 (0.451,0.511)	0.259 (0.235,0.282)	0.097 (0.086,0.107)	0.119 (0.106,0.132)	0.045 (0.040,0.051)

^a Estimations obtained omitting judgements $a_{12}^{(3)}, a_{35}^{(3)}, a_{34}^{(4)}$ and $a_{45}^{(4)}$.

^b Estimations obtained making judgements $a_{12}^{(3)}, a_{35}^{(3)}, a_{34}^{(4)}$ and $a_{45}^{(4)}$ imprecise.

Table 5.5
Levels of consistency estimated using each prioritization method (Bayesian credibility interval of 95% in parentheses)

	$\sigma^{(1)}$	$\sigma^{(2)}$	$\sigma^{(3)}$	$\sigma^{(4)}$	$\sigma^{(5)}$	$\sigma^{(6)}$	MSE
AIJ and AIP	0.348 (0.235,0.590)	0.636 (0.430,1.079)	0.861 (0.581,1.460)	0.897 (0.606,1.522)	0.602 (0.406,1.021)	0.218 (0.147,0.371)	0.545
Bayesian	0.278 (0.199,0.422)	0.518 (0.377,0.771)	0.868 (0.633,1.284)	0.905 (0.659,1.345)	0.588 (0.429,0.872)	0.128 (0.088,0.211)	0.504
AIJ ^a	0.296 (0.200,0.503)	0.584 (0.395,0.991)	0.795 (0.514,1.459)	0.744 (0.482,1.367)	0.635 (0.429,1.078)	0.201 (0.136,0.341)	0.500
AIP ^a	0.310 (0.209,0.525)	0.596 (0.403,1.012)	0.695 (0.450,1.276)	0.633 (0.410,1.162)	0.623 (0.421,1.057)	0.196 (0.132,0.333)	0.447
Bayesian ^a	0.278 (0.200,0.420)	0.519 (0.378,0.769)	0.643 (0.468,0.949)	0.593 (0.430,0.882)	0.587 (0.429,0.868)	0.128 (0.087,0.212)	0.328
Bayesian ^b	0.278 (0.200,0.420)	0.519 (0.378,0.769)	0.681 (0.493,1.005)	0.728 (0.524,1.087)	0.586 (0.429,0.871)	0.128 (0.088,0.213)	0.328

^a Estimations obtained omitting judgements $a_{12}^{(3)}, a_{35}^{(3)}, a_{34}^{(4)}$ and $a_{45}^{(4)}$.

^b Estimations obtained making judgements $a_{12}^{(3)}, a_{35}^{(3)}, a_{34}^{(4)}$ and $a_{45}^{(4)}$ imprecise.

Table 5.6

Matrices of domination between alternatives for each priority estimation method

	A_1	A_2	A_3	A_4	A_5
AIJ					
A_1	0.00	1.00	1.00	1.00	1.00
A_2	0.00	0.00	1.00	1.00	1.00
A_3	0.00	0.00	0.00	0.02	1.00
A_4	0.00	0.00	0.98	0.00	1.00
A_5	0.00	0.00	0.00	0.00	0.00
AIP					
A_1	0.00	1.00	1.00	1.00	1.00
A_2	0.00	0.00	1.00	1.00	1.00
A_3	0.00	0.00	0.00	0.25	1.00
A_4	0.00	0.00	0.75	0.00	1.00
A_5	0.00	0.00	0.00	0.00	0.00
Bayesian					
A_1	0.00	1.00	1.00	1.00	1.00
A_2	0.00	0.00	1.00	1.00	1.00
A_3	0.00	0.00	0.00	0.01	1.00
A_4	0.00	0.00	0.99	0.00	1.00
A_5	0.00	0.00	0.00	0.00	0.00

Table 5.7

Most likely preference structures for each priority estimation method

	12345	12435	12453	14235	21435	21345
AIJ	1.30	98.70	0.00	0.00	0.00	0.00
AIP	24.50	75.22	0.05	0.02	0.15	0.06
Bayesian	1.35	98.63	0.01	0.01	0.00	0.00
AIJ ^a	0.38	99.62	0.00	0.01	0.00	0.00
AIP ^a	13.63	86.27	0.01	0.06	0.02	0.01
Bayesian ^a	1.20	98.80	0.00	0.00	0.00	0.00
Bayesian ^b	1.40	98.60	0.00	0.00	0.00	0.00

^a Estimations obtained omitting judgements $a_{12}^{(3)}$, $a_{35}^{(3)}$, $a_{34}^{(4)}$ and $a_{45}^{(4)}$.^b Estimations obtained making judgements $a_{12}^{(3)}$, $a_{35}^{(3)}$, $a_{34}^{(4)}$ and $a_{45}^{(4)}$ imprecise.

Essentially, the results obtained from all of the methods appear to coincide, with all of them giving rise to the group preference structure [12435]. This is so, on the one hand, because it is preferred by the majority of decision makers and, on the other, because of the high inconsistency indices for the decision makers who are most in disagreement, implying that these have less weight in the combined estimation of the group priority vector using Bayesian methods.

The Bayesian estimates best reflect the opinions of the group, as shown by the smaller mean quadratic error, as well as the lower levels of inconsistency estimated for the majority of the decision makers (see Table 5.5), and, above all, for the most consistent decision makers (D_1 and D_6).

The differences in the levels of inconsistency for the remaining decision makers estimated by using the AIP and AIJ methods are not significant. This translates into a lower level of uncertainty in the estimation of the group's priorities, resulting in more accurate credibility intervals (see Table 5.4), probabilities of rank reversal that are closer to 0 and 1 (see Table 5.6), and a more concentrated posterior distribution of preference structures.

5.3. Incomplete information

In order to analyse the reasons for the disagreements identified in the above analysis, this section describes, by way of example, a study of the influence of certain judgements on the estimated consensus

Table 5.8
Variability of judgements expressed by decision makers^a

Judgement	D_1	D_2	D_3	D_4	D_5	D_6	Opening coefficient ^b
a_{12}	3	4	1/2	3	2	2	8
a_{13}	5	3	3	5	6	5	5/3
a_{14}	4	5	2	2	3	4	5/2
a_{15}	7	8	5	6	3	9	3
a_{23}	3	4	5	1	2	3	5
a_{24}	2	3	1	3	5	2	5
a_{25}	5	6	2	2	4	6	3
a_{34}	1/2	1	2	4	1/2	1	8
a_{35}	3	5	1/2	5	1	2	10
a_{45}	3	7	5	1/2	5	3	14

^a The judgements omitted from the analysis in Section 4.1 are shown in bold type.

^b The opening coefficient is given by: $\text{Max}_k a_{ij}^{(k)} / \text{Min}_k a_{ij}^{(k)}, k = 1, \dots, 6$.

priorities. Table 5.8 shows the judgements expressed by the decision makers, together with a measure of their variability (opening coefficient). Note that the highest levels of variability are found in judgements, a_{34} , a_{35} and a_{45} , and, in particular, that the judgements expressed by decision makers D_3 (judgements $a_{12}^{(3)}$ and $a_{35}^{(3)}$) and D_4 (judgements $a_{34}^{(4)}$ and $a_{45}^{(4)}$) are those that disagree most with the preference structure [12435], which is the one preferred by the other decision makers (see Table 5.3). Consequently, we shall repeat the above analysis omitting these judgements.

As Table 5.5 shows, the omission of these judgements raises the consistency levels for decision makers D_3 and D_4 , and also increases the degree of consensus among all of the decision makers with the disappearance of D_3 's disagreement with the rest of the group. However, the discrepancies of decision maker D_4 persist, on the one hand, because of the high probability of rank reversal between A_2 and A_3 ($P(A_3 > A_2 | y^{(4)}) = 32\%$) and, on the other, because of the total discrepancy in the relative position of alternatives A_3 and A_4 ($P(A_3 > A_4 | y^{(4)}) = 2\%$) with regard to the rest of the decision makers.

5.4. Imprecise information

Another way of trying to improve consensus levels is to weaken the degree of precision in the expression of the above judgements, replacing them with intervals that increase the degree of flexibility for the most discordant decision makers. In this case, given that the positions of decision makers D_3 and D_4 are the most discordant, we substitute the precise information available with regard to judgements $a_{12}^{(3)}$, $a_{35}^{(3)}$, $a_{34}^{(4)}$ and $a_{45}^{(4)}$ for the intervals $[1/3, 1]$, $[1/3, 1]$, $[3, 5]$ and $[1/3, 1]$, respectively. The results obtained from the application of the algorithm described in Appendices A and B over 10,000 iterations, discarding the first 5000, are shown in Tables 5.5 and 5.7. The estimates obtained for priorities are practically identical to those obtained on eliminating the judgements, but with a slight rise in the level of inconsistency estimated for decision makers D_3 and D_4 , due to the restrictions imposed on those judgements.

6. Conclusions

In this paper, we have proposed, analysed and compared (in terms of consistency) three procedures for estimating local priorities in AHP-group decision making. Two of these procedures are conventionally used in the AHP literature: the aggregation of the individual judgements (AIJ) and the aggregation of individual priorities (AIP). The third procedure corresponds to our proposal, the Bayesian prioritization procedure (BPP). This takes the form of a Bayesian method that does not require intermediate filters for the initial judgements of the actors. It is based on a Bayesian analysis of the problem using a multiplicative model with log-normal errors, which describes the valuation process under the hypothesis of the existence of consensus among the decision makers.

BPP is essentially a process of weighted aggregation of individual priorities using weightings that are inversely proportional to the decision makers' levels of inconsistency. This method is more efficient than the

estimation of consensus priorities using the two conventional techniques. It also naturally extends to the case of incomplete and imprecise pairwise comparison matrices, where it performs in a notably more robust manner than the conventional methods, especially with regard to consistency. It also allows for the inclusion of the uncertainty associated with the priorities estimation process in the analysis of individual preference structures, as well as the possible existence of a change of rank between alternatives. For these reasons, we would argue that the proposed BPP prioritization method is more flexible, practical and realistic than the other two procedures with which it has been compared.

This study has been based on the prior assumption of the existence of consensus among the agents implicated in the problem. We are currently working on an extension of the methodology to locate and identify areas of partial consensus between sub-groups of decision makers. This information would be provided to the actors involved in the resolution of the problem to enable them to establish consensus paths in subsequent negotiation processes. The results of this line of research, as well as those corresponding to the extension of the study to informative priors, will be described in subsequent papers.

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Appendix A

Proof of Theorem 1. From (3.2) it follows that the posterior distribution of the vector will be given by:

$$\begin{aligned} [\mu^G | y] &\propto [y | \mu^G] [\mu^G] \propto \prod_{k=1}^r [y^{(k)} | \mu^G] [\mu^G] \propto \prod_{k=1}^r \exp \left[-\frac{\tau^{(k)}}{2} (y^{(k)} - X\mu^G)' (y^{(k)} - X\mu^G) \right] \\ &\propto \exp \left[-\frac{1}{2} \mu^{G'} \left(\sum_{k=1}^r \tau^{(k)} (X'X) \right) \mu^G + \mu^{G'} \left(\sum_{k=1}^r \tau^{(k)} (X'y^{(k)}) \right) \right]. \end{aligned} \quad (A.1)$$

From (A.1) it follows that:

$$\mu^G | y \sim N_{n-1}(\hat{\mu}_B, \hat{\Sigma}_B), \quad (A.2)$$

where $\hat{\mu}_B = (\sum_{k=1}^r \tau^{(k)} (X'X))^{-1} (\sum_{k=1}^r \tau^{(k)} (X'y^{(k)}))$ and $\hat{\Sigma}_B = (\sum_{k=1}^r \tau^{(k)} (X'X))^{-1}$.

Taking into account that $\hat{\mu}^{(k)} = (X'X)^{-1} (X'y^{(k)})$, the theorem follows. $\square$

Proof of Theorem 2. It can be observed that the estimators $\hat{\mu}_{AIJ}$ and $\hat{\mu}_B$ are unbiased for μ^G and that their variance and covariance matrices are given, respectively, by

$$\text{Var}(\hat{\mu}_{AIJ}) = \frac{1}{r^2} \left(\sum_{k=1}^r \sigma^{(k)2} \right) (X'X)^{-1} \quad \text{and} \quad \text{Var}(\hat{\mu}_B) = \frac{1}{(\sum_{k=1}^r \frac{1}{\sigma^{(k)2}})} (X'X)^{-1}.$$

The difference is given by

$$\text{Var}(\hat{\mu}_{AIJ}) - \text{Var}(\hat{\mu}_B) = \frac{1}{r} \left\{ \frac{1}{r} \sum_{k=1}^r \sigma^{(k)2} - \frac{r}{(\sum_{k=1}^r \frac{1}{\sigma^{(k)2}})} \right\} (X'X)^{-1} = \frac{1}{r} \{ \bar{\sigma}_A^2 - \bar{\sigma}_H^2 \} (X'X)^{-1}, \quad (A.3)$$

where $\bar{\sigma}_A^2$ and $\bar{\sigma}_H^2$ represent, respectively, the arithmetic and harmonic mean of the variances $\{\sigma^{(k)2}, k = 1, \dots, r\}$. It follows from (A.3) that $\text{Var}(\hat{\mu}_{AIJ}) - \text{Var}(\hat{\mu}_B)$ is a positive semi-definite matrix, because $X'X$ is similarly one and because that $\bar{\sigma}_A^2 \geq \bar{\sigma}_H^2$. Hence, the Bayesian estimator $\hat{\mu}_B$ is more efficient than the estimator based on the unweighted aggregation of judgements $\hat{\mu}_{AIJ}$. Moreover, the Bayesian estimator $\hat{\mu}_B$ is also the GLS estimator of μ^G under the normal linear model (3.1), which is well known to be the most efficient of the unbiased estimators of μ^G (Judge et al., 1985, p. 27). $\square$

Proof of Theorem 4. If $\hat{\mu} = \hat{\mu}_{AIJ}$ or $\hat{\mu}_{AIP}$, then it is verified that:

$$\text{Var}(\hat{\mu}) = \text{Var}(\hat{\mu} - \hat{\mu}_B + \hat{\mu}_B) = \text{Cov}(\hat{\mu}_B, \hat{\mu} - \hat{\mu}_B) + \text{Cov}(\hat{\mu} - \hat{\mu}_B, \hat{\mu}_B) + \text{Var}(\hat{\mu}_B) + \text{Var}(\hat{\mu} - \hat{\mu}_B). \quad (\text{A.4})$$

However, as $\hat{\mu}_B = \mathbf{E}_B \mathbf{y}$ with $\mathbf{E}_B = \{(\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) \mathbf{X}' \mathbf{D}^{-1} \mathbf{X} (\mathbf{1}^{\otimes} \mathbf{I}_{n-1})\}^{-1} (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) \mathbf{X}' \mathbf{D}^{-1}$, it follows that:

$$\begin{aligned} \text{Cov}(\hat{\mu}_B, \hat{\mu} - \hat{\mu}_B) &= \mathbf{E}_B \text{Var}(\mathbf{y})(\mathbf{E} - \mathbf{E}_B)' = \mathbf{E}_B \mathbf{D}(\mathbf{E} - \mathbf{E}_B)' \\ &= \{(\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) \mathbf{X}' \mathbf{D}^{-1} \mathbf{X} (\mathbf{1}^{\otimes} \mathbf{I}_{n-1})\}^{-1} (\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) \mathbf{X}' (\mathbf{E}' - \mathbf{E}'_B) \\ &= \{(\mathbf{1}'_r \otimes \mathbf{I}_{n-1}) \mathbf{X}' \mathbf{D}^{-1} \mathbf{X} (\mathbf{1}^{\otimes} \mathbf{I}_{n-1})\}^{-1} (\mathbf{I}_{n-1} - \mathbf{I}_{n-1}) = \mathbf{0}. \end{aligned}$$

Hence, from (A.4) we have

$$\text{Var}(\hat{\mu}) = \text{Var}(\hat{\mu}_B) + \text{Var}(\hat{\mu} - \hat{\mu}_B)$$

and $\hat{\mu}_B$ will, therefore, be more efficient than $\hat{\mu}$. In particular, both estimators $\hat{\mu} = \hat{\mu}_{AIJ}$ and $\hat{\mu} = \hat{\mu}_{AIP}$ will be less efficient than $\hat{\mu}_B$. $\square$

Appendix B

This Appendix B describes an algorithm that allows us to make inferences about the vector of log-priorities μ^G and the variances $\{\sigma^{(k)2}; k = 1, \dots, r\}$ of model (4.2) when such variances are unknown and the available information is incomplete and/or imprecise. The algorithm is based on the application of the *Gibbs sampling* technique to the posterior distribution (4.9), the full conditional distributions of which are standard and are given by the algorithm itself. The convergence of the algorithm follows from Lemma 7.2.1 of Robert and Casella (1999).

Algorithm (Extension):

Step 0 (Initialisation):

Complete the vectors of the judgements expressed by each decision maker $\mathbf{y}^{(k,0)}$, assigning initial values to the imprecise judgements using, for example, simulation techniques for interval judgements. Also assign initial values, $\{\tau^{(k,0)}; k = 1, \dots, r\}$, to the precision parameters $\{\tau^{(k)}; k = 1, \dots, r\}$. Set $it = 1$ and repeat Steps 1–2 below n_{iter} times.

Step 1 (Sampling of the full conditional distribution of μ^G):

Extract $\mu^{G(it)}$ from $N_{n-1}(\hat{\mu}_B^{(it-1)}, \sum_{k=1}^r \tau^{(k,it-1)} (\mathbf{X}^{(k)'} \mathbf{X}^{(k)})^{-1})$ with

$$\hat{\mu}_B^{(it-1)} = \left(\sum_{k=1}^r \tau^{(k,it-1)} (\mathbf{X}^{(k)'} \mathbf{X}^{(k)}) \right)^{-1} \left(\sum_{k=1}^r \tau^{(k,it-1)} (\mathbf{X}^{(k)'} \mathbf{y}^{(k,it-1)}) \right).$$

Step 2 (Sampling of the full conditional distribution of $\{\tau^{(k)}; k = 1, \dots, r\}$):

Draw $\tau^{(k,it)}$ of Gamma $\left(\frac{J_k}{2}, \frac{(\mathbf{y}^{(k,it-1)} - \mathbf{X}^{(k)} \mu^{G(it)})' (\mathbf{y}^{(k,it-1)} - \mathbf{X}^{(k)} \mu^{G(it)})}{2} \right)$ for $k = 1, \dots, r$, where $\mathbf{y}^{(k,it-1)}$ contains the judgements expressed by decision maker $\mathbf{D}_k$ with the imprecise judgements assigned in the previous iteration.

Step 3 (Sampling of imprecise judgements):

For each imprecise judgement $y_{ij}^{(k)}$ assign a value of $y_{ij}^{(k,it)}$ drawing a sample from the normal distribution $N(\mu_i^{(k,it)} - \mu_j^{(k,it)}, \sigma^{(k,it)2})$ truncated at the interval $[y_{ij}^{(k)}, \bar{y}_{ij}^{(k)}]$, where $\sigma^{(k,it)2} = \frac{1}{\tau^{(k,it)}}$. On the basis of these imputations, construct the judgement vectors $\{\mathbf{y}^{(k,it)}; k = 1, \dots, r\}$.

After a “burn-in” period of n_{iter0} to achieve convergence, this algorithm will provide a sample

$$\{(\mu^G(it), \sigma^{(k,it)2}, \mathbf{y}^{(k,it)}); k = 1, \dots, r; it = n_{\text{iter0}} + 1, \dots, n_{\text{iter}}\},$$

which may be used to make inferences about the priorities of the group μ^G , as well as the decision makers’ degrees of consistency.

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